



جامعة العلوم الإسلامية العالمية
The World Islamic Sciences and Education University



كلية تكنولوجيا المعلومات
Faculty of Information Technology

Course: System Analysis and Design (0604334)

Doctor: Moath Tarawneh

Date: 16/4/2026

Team Members:

- | | |
|------------------|----------------------|
| 1. Joud Kayyali | 2. Zaid Alassaf |
| 3. Hamzah Altoom | 4. Ibraheem Abdullah |

PrintSmart: Unified Print Estimation & Inventory System

Phase 1: Systems Proposal & Project Charter

1. Problem Definition

The current pricing process within the print shop relies heavily on manual calculations and estimations. Employees utilize fixed Excel sheets that are incapable of covering custom print orders, which forces them to frequently consult the manager for accurate pricing. This dependency creates an operational bottleneck, wastes valuable time, and ultimately leads to inaccurate or inconsistent pricing. The core disparity lies in information accessibility: employee's base prices on worst-case scenarios (e.g., maximum waste, bleed, margins, and full-color usage) using static data. Conversely, the owner has exclusive access to granular operational metrics, including machine wattage

consumption, fluctuating material costs, maintenance expenses, rent, and distinct B2B versus B2C pricing structures. Consequently, employees must repeatedly seek the owner's input to offer competitive prices and close sales.

2. Identified Issues

The manual and fragmented nature of the current workflow yields several critical issues:

- Reliance on fixed, static pricing that fails to accommodate dynamic print requests.
- Frequent operational delays as employees seek managerial approval for cases not covered in the static pricing sheets.
- Inability to dynamically calculate B2B pricing, resulting in a failure to offer customized rates for loyal clients.
- Lack of transparency regarding the actual cost of orders for non-administrative employees.
- High cognitive load on employees, who must memorize material constraints, machine specifics, and potential error margins before quoting a price.
- Customer dissatisfaction and confusion arising from receiving different price quotes for the same service across different visits.
- Loss of competitive advantage due to inaccurate and rigid pricing models.

- Extensive retraining and manual system updates required whenever a new printer is acquired or material compatibility changes.
- Absence of a systemic approach to inventory tracking, leaving stock assessment to individual employee estimation.

3. Project Objectives

The primary objectives of the PrintSmart system are to:

- Automate the pricing process to cover all dynamic custom cases without manager intervention.
- Calculate dynamic pricing accurately based on specific request details, including errors, waste, and cutting margins.
- Incorporate a tiered pricing system to accommodate B2B clients and frequent customers.
- Secure and conceal actual cost data and sensitive financial metrics from non-administrative staff.
- Provide an intuitive, easy-to-use interface that ensures pricing clarity for both employees and customers.
- Scale efficiently to allow the configuration of new printers and evolving pricing criteria.
- Ensure compliance with local legal and tax regulations in Jordan.
- Maintain real-time and accurate inventory tracking to prevent material shortages.

- Manage edge cases automatically, such as machine failures and human errors, by adjusting calculations based on specific machine and material details.
- Provide decision-makers with accurate reports to evaluate business performance and optimize future operations.

4. Project Scope

- **In-Scope:** The proposed PrintSmart system functions strictly as a dynamic price estimation and inventory tracking tool.
- **Out-of-Scope:** This project explicitly excludes external accounting systems, invoice generation systems, and sales/marketing modules.

5. Stakeholders Analysis

- **Primary Users (In-Scope):**
 - **Front Desk Employee:** Handles customer sales, quick paper prints, and acts as the main user of the new system.
 - **Graphic Designer:** Prepares customer designs to be compatible with printing machines and formats.
 - **Machine Operator:** Operates various machines (laser, CNC, UV, etc.) and performs fundamental maintenance.
 - **Supervisor (Order Coordinator):** Manages the team and resolves edge cases as a sub-admin.

- **Manager/Owner:** Handles strategic planning, B2B deals, and holds full admin privileges. (Note: Currently, two employees cover multiple operational roles alongside the manager, and there is no dedicated stock manager).
- **Excluded Actors (Out-of-Scope):**
 - Outsourced Marketing Team.
 - Outsourced Developers.
 - Human Resources (HR).

6. System Requirements

6.1. Functional Requirements:

- The system must calculate the final price depending on materials, machines, services, waste, bleed, margin, and specific order details (e.g., color usage and non-printing area).
- The system must apply special pricing tiers for loyal customers and B2B clients.
- The system must automatically deduct materials from the inventory once a customer confirms an order.
- The system must prohibit non-administrative users from viewing the actual costs of materials and operations.
- The system must generate and output a detailed, dated Quotation Price for the customer.

6.2. Non-Functional Requirements:

- **Usability:** The system interface should be intuitive enough that a new employee can utilize it efficiently with less than 2 hours of training.
- **Security:** The system must use Role-Based Access Control (RBAC) to ensure only the Manager can view material costs.
- **Performance:** The system must calculate and return the final dynamic price in less than 3 seconds.
- **Platform:** The system must be Web-based to support BYOD (Bring Your Own Device) policies for employees and managers.

7. Methodology Used to Perform the Project

Based on the project requirements and the need for a scalable, well-documented system, we will adopt a **Hybrid Methodology** that separates Project Management from System Engineering:

1. **Project Management Level (SDLC):** For the high-level management view, timeline tracking, and feasibility, we will strictly follow the Systems Development Life Cycle (SDLC). This phase (Phase 1 & 2) includes submitting the project proposal, defining objectives, and utilizing management tools such as Network diagrams, PERT, and Gantt charts to identify critical paths and ensure timely delivery. Establishing this uniform management structure is a prerequisite before handling any large-scale changes or future scalability.
2. **Systems Analysis & Design Level (Modular/Structured to O-O):** Once the SDLC

planning is established, the engineering phase will adopt a Modular Design Strategy.

- *Current Analysis (Phase 2 Focus):* To accurately map the complex logic of the print shop, we will use Structured Analysis tools, specifically Data Flow Diagrams (DFDs) and Event Response Tables, divided by core modules: (1) Cost Estimation/Sales, (2) Machine Operations, and (3) Inventory Management.
- *Future Implementation (O-O Strategy):* These structured modules will act as a foundation for an Object-Oriented (O-O) implementation. Each module will eventually be decomposed into specific O-O Classes and Use Cases. This hybrid separation ensures that the analysis is deeply detailed while providing a robust, class-based blueprint for future development and deployment.

8. Function Points for Estimation

To estimate the size and complexity of the PrintSmart system, Function Point Analysis (FPA) was conducted across the core system transactions.

8.1. Unadjusted Function Points (UFP) Calculation:

Component Type	Description	Complexity	Points
Inputs	Screen input for order details and pricing	Complex	6

Component Type	Description	Complexity	Points
	Screen input for customer details	Simple	2
	Screen input for Inventory and stock	Average	4
	Screen input for machines and pricing configurations	Complex	6
	Inputs Subtotal		18
Outputs	Invoice or quotation price for customers	Complex	10
	Shortage in stock	Simple	4
	Report for waste and profit	Average	6
	Outputs Subtotal		20
Files (Tables)	Stock and Inventory table	Simple	2
	Machines table	Average	4
	Customers table	Simple	2
	Orders and print operations table	Complex	8
	Services table	Simple	2

Component Type	Description	Complexity	Points
	Files Subtotal		18
Inquiries	Search and filtering for stock	Average	10
	Search on orders	Average	10
	Inquiries Subtotal		20
Interfaces	No external APIs currently in scope	N/A	0
Total UFP			76 points

8.2. Adjusted Function Point Calculation: General System Characteristics (GSCs) were evaluated to determine the Project Adjustment Factor:

- Transaction Rate: 2
- Complex Processing: 5
- Performance: 3
- Code to be reusable: 3
- Data Communications: 3
- Facilitate Change: 4
- Operational Ease: 4
- End-User Efficiency: 5
- Online Update: 3

- Heavily Used Configuration: 3
- **Total Project Adjustment Factor = 35**

Final Formula Application:

- Adjusted FP = Unadjusted FP × [0.65 + (Adjustment Factor × 0.01)]
- Adjusted FP = 76 × [0.65 + (35 × 0.01)]
- Adjusted FP = 76 × 1.00 = **76 Adjusted Function Points**

Estimation Conclusion: Assuming an organizational average productivity rate of 18 function points per month, the estimated development effort is calculated as: **76 FP / 18 = 4.2 Man-Months.**